

## CIUDAD DEL ESTE GUARANI, Paraguay

WMO: 862183

Lat: 25.450S

Lon: 54.833W

Elev: 264

StdP: 98.19

Time Zone: -4.00 (PRY)

Period: 98-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 5.0        | 7.0 | 1.8                           | 4.4 | 9.2  | 3.2 | 4.9 | 9.8  | 10.4                  | 21.3 | 9.4 | 22.7 | 2.2                   | 180  | 0.362 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 10.7                   | 36.1            | 23.8 | 35.1 | 23.8 | 34.0 | 23.9 | 26.7                | 32.1 | 26.1 | 31.5 | 25.6 | 30.9 | 4.9                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.2                            | 21.0 | 29.6 | 24.8 | 20.4 | 29.1 | 24.1 | 19.6 | 28.2 | 85.7          | 32.1 | 82.5 | 31.9 | 80.2 | 31.4 | 30.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 9.5               | 8.3  | 7.4 | 1.7                        | 38.6 | 1.4                | 0.8 | 0.6                                                | 39.2 | -0.2       | 39.6 | -1.0       | 40.1 | -2.0       | 40.6 |
|                   |      |     | 0.7                        | 28.3 | 1.5                | 0.9 | -0.4                                               | 29.0 | -1.2       | 29.5 | -2.0       | 30.0 | -3.1       | 30.7 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 22.7   | 26.9  | 26.7  | 25.8  | 23.3  | 19.1  | 18.0  | 17.5  | 19.5  | 21.2  | 23.8  | 24.7  | 26.1  |
|                                                                        | DBStd       | 4.82   | 1.81  | 1.97  | 2.35  | 3.41  | 3.60  | 4.23  | 4.56  | 4.88  | 4.63  | 3.22  | 2.58  | 2.25  |
|                                                                        | HDD10.0     | 7      | 0     | 0     | 0     | 0     | 1     | 2     | 3     | 2     | 0     | 0     | 0     | 0     |
|                                                                        | HDD18.3     | 243    | 0     | 0     | 0     | 6     | 34    | 58    | 71    | 48    | 24    | 1     | 0     | 0     |
|                                                                        | CDD10.0     | 4642   | 526   | 468   | 491   | 398   | 282   | 242   | 237   | 296   | 336   | 429   | 439   | 498   |
|                                                                        | CDD18.3     | 1836   | 267   | 235   | 233   | 154   | 58    | 48    | 47    | 84    | 109   | 172   | 190   | 239   |
|                                                                        | CDH23.3     | 16528  | 2638  | 2214  | 2107  | 1196  | 337   | 265   | 326   | 745   | 1054  | 1555  | 1798  | 2292  |
| (13)                                                                   | CDH26.7     | 7133   | 1214  | 1016  | 940   | 458   | 83    | 52    | 78    | 292   | 485   | 703   | 789   | 1021  |
| (14)                                                                   | Wind        | WSAvg  | 3.3   | 3.1   | 3.0   | 3.0   | 3.0   | 3.1   | 3.3   | 3.8   | 3.9   | 3.8   | 3.5   | 3.3   |
| (15) Precipitation                                                     | PrecAvg     | 1742   | 180   | 148   | 128   | 150   | 163   | 106   | 100   | 101   | 138   | 178   | 190   | 178   |
|                                                                        | PrecMax     | 2468   | 383   | 401   | 347   | 297   | 491   | 318   | 301   | 346   | 321   | 282   | 371   | 459   |
|                                                                        | PrecMin     | 1203   | 63    | 0     | 26    | 61    | 0     | 1     | 13    | 30    | 19    | 9     | 59    | 0     |
|                                                                        | PrecStd     | 374    | 70    | 97    | 92    | 57    | 119   | 78    | 69    | 73    | 84    | 76    | 91    | 111   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 36.3  | 37.2  | 36.2  | 34.4  | 31.1  | 29.9  | 30.2  | 34.0  | 37.0  | 37.8  | 36.2  | 36.2  |
|                                                                        |             | MCWB   | 24.7  | 24.5  | 23.9  | 24.4  | 22.2  | 20.8  | 20.1  | 20.9  | 22.2  | 24.4  | 23.7  | 24.2  |
|                                                                        | 2%          | DB     | 34.9  | 35.2  | 34.8  | 32.9  | 29.0  | 28.1  | 28.9  | 31.9  | 34.8  | 35.1  | 34.2  | 34.8  |
|                                                                        |             | MCWB   | 24.7  | 24.6  | 24.0  | 23.7  | 21.9  | 20.6  | 19.8  | 20.2  | 21.0  | 23.5  | 23.3  | 24.3  |
|                                                                        | 5%          | DB     | 33.8  | 33.8  | 33.1  | 31.2  | 27.1  | 26.8  | 27.1  | 30.0  | 32.1  | 33.0  | 32.9  | 33.2  |
|                                                                        |             | MCWB   | 24.7  | 24.6  | 23.9  | 23.2  | 20.9  | 20.2  | 19.7  | 19.8  | 20.9  | 22.7  | 23.1  | 24.3  |
|                                                                        | 10%         | DB     | 32.2  | 32.1  | 31.8  | 29.8  | 25.0  | 24.4  | 25.0  | 27.9  | 29.2  | 30.8  | 31.1  | 31.9  |
|                                                                        |             | MCWB   | 24.5  | 24.0  | 23.6  | 22.8  | 20.0  | 19.7  | 18.9  | 19.5  | 19.9  | 22.0  | 22.6  | 23.9  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 27.6  | 27.3  | 26.6  | 26.4  | 24.3  | 22.7  | 22.2  | 23.0  | 24.2  | 26.2  | 26.9  | 27.1  |
|                                                                        |             | MCDB   | 32.5  | 32.9  | 31.9  | 31.6  | 28.4  | 26.0  | 27.2  | 29.9  | 32.8  | 34.2  | 32.4  | 32.5  |
|                                                                        | 2%          | WB     | 26.6  | 26.2  | 25.6  | 25.0  | 22.9  | 21.5  | 21.1  | 21.7  | 22.9  | 24.8  | 25.5  | 26.3  |
|                                                                        |             | MCDB   | 31.9  | 31.7  | 31.1  | 29.9  | 26.7  | 25.7  | 26.5  | 28.2  | 30.7  | 31.9  | 31.2  | 31.4  |
|                                                                        | 5%          | WB     | 25.9  | 25.6  | 25.0  | 24.2  | 21.9  | 20.7  | 20.1  | 20.7  | 21.8  | 23.9  | 24.5  | 25.6  |
|                                                                        |             | MCDB   | 31.2  | 30.9  | 30.4  | 28.8  | 25.2  | 25.0  | 25.1  | 27.5  | 28.1  | 30.4  | 30.0  | 30.6  |
|                                                                        | 10%         | WB     | 25.4  | 25.0  | 24.4  | 23.3  | 20.9  | 19.9  | 19.3  | 19.8  | 20.9  | 23.0  | 23.7  | 24.9  |
|                                                                        |             | MCDB   | 30.4  | 30.1  | 29.5  | 27.9  | 24.0  | 23.5  | 23.7  | 26.1  | 27.0  | 28.8  | 28.7  | 29.6  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 10.7  | 10.7  | 10.9  | 10.5  | 9.7   | 9.8   | 10.9  | 12.3  | 12.2  | 11.3  | 11.4  | 10.8  |
|                                                                        |             | MCDBR  | 12.7  | 13.0  | 13.0  | 12.3  | 11.8  | 11.7  | 13.3  | 14.7  | 15.7  | 14.5  | 13.8  | 12.7  |
|                                                                        |             | MCWBR  | 4.6   | 4.7   | 4.6   | 4.8   | 5.6   | 5.4   | 6.0   | 5.9   | 5.7   | 5.5   | 5.4   | 5.0   |
|                                                                        | 5% WB       | MCDBR  | 10.6  | 10.8  | 10.8  | 10.6  | 9.8   | 10.3  | 11.7  | 13.1  | 13.1  | 12.5  | 11.4  | 10.6  |
|                                                                        |             | MCWBR  | 4.9   | 4.8   | 4.8   | 4.9   | 5.4   | 5.5   | 6.1   | 6.3   | 6.0   | 5.5   | 5.5   | 5.1   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.402 | 0.397 | 0.389 | 0.384 | 0.358 | 0.388 | 0.387 | 0.469 | 0.566 | 0.475 | 0.417 | 0.414 |
|                                                                        | taud        |        | 2.425 | 2.438 | 2.453 | 2.437 | 2.479 | 2.391 | 2.371 | 2.096 | 1.851 | 2.171 | 2.339 | 2.369 |
|                                                                        | Ebn at Noon |        | 942   | 931   | 910   | 867   | 849   | 791   | 807   | 769   | 737   | 850   | 922   | 932   |
|                                                                        | Edh at Noon |        | 125   | 120   | 113   | 105   | 92    | 96    | 101   | 145   | 203   | 155   | 135   | 132   |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 6.46  | 5.92  | 5.40  | 4.44  | 3.29  | 2.85  | 3.22  | 3.96  | 4.53  | 5.19  | 6.30  | 6.50  |
|                                                                        | RadStd      |        | 0.51  | 0.47  | 0.36  | 0.41  | 0.46  | 0.30  | 0.38  | 0.29  | 0.49  | 0.59  | 0.60  | 0.59  |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (3 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page